Data-Driven Threshold Optimization in Consensus Machine Learning for High-Imbalance Toxicity Prediction: A Tox21 Case Study

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Abstract

Predicting in vivo toxicity from in vitro high-throughput screening (qHTS) data remains a critical bottleneck in computational toxicology. The Tox21 dataset presents significant challenges due to extreme class imbalance and high-dimensional feature noise. In this study, we propose an optimized consensus machine learning pipeline designed to predict toxicity across 12 distinct biological endpoints. Utilizing ECFP4 molecular fingerprints, we applied a dual-stage feature selection methodology (Variance Thresholding and Random Forest importance) to isolate biologically relevant substructures, reducing the feature space by approximately 79%. We deployed a consensus model combining Bernoulli Naïve Bayes and XGBoost algorithms, weighted by their respective ROC-AUC performance. Crucially, we abandoned static binary classification thresholds (e.g., 0.50) in favor of dynamic, data-driven thresholds optimized strictly for the Matthews Correlation Coefficient (MCC). This approach yielded highly specific, pathway-dependent models capable of accurately isolating mechanisms of action—successfully identifying Bisphenol A as a precise Estrogen Receptor (ER) disruptor and Benzo[a]pyrene as an Aryl Hydrocarbon Receptor (AhR) agonist, while maintaining zero false positives for negative controls.

Keywords: Predictive Toxicology, Tox21, Machine Learning, XGBoost, Consensus Modeling, Class Imbalance, MCC Optimization

1 Introduction

The Toxicology in the 21st Century (Tox21) program provides an unprecedented volume of in vitro biological assay data, evaluating thousands of environmental chemicals and drugs across key nuclear receptor and stress response pathways. Despite the availability of this data, standard machine learning models frequently exhibit poor predictive utility on the minority (toxic) class due to severe class imbalance. Models evaluated on standard accuracy metrics routinely overfit to the majority (non-toxic) class.

To overcome these limitations, this study introduces a highly constrained, data-driven consensus modeling approach. By combining robust feature selection, algorithmic ensemble weighting, and strict MCC-optimized dynamic thresholding, we aim to build highly specific predictive models for all 12 Tox21 endpoints that minimize false discovery rates without sacrificing minority class sensitivity.

2 Methodology

2.1 Data Preparation and Featurization

Data was sourced from the Tox21 dataset via the MoleculeNet repository. Chemical structures, provided as Simplified Molecular-Input Line-Entry System (SMILES) strings, were translated into Extended-Connectivity Fingerprints (ECFP4) with a radius of 2 and a length of 1024 bits using RDKit. Compounds with invalid chemical structures were systematically removed prior to training.

2.2 Dual-Stage Feature Selection

To mitigate the curse of dimensionality and remove uninformative zero-variance features inherent to 1024-bit fingerprints, a two-step feature selection pipeline was implemented independently for each of the 12 biological endpoints:

- **Variance Thresholding:** Features exhibiting a variance lower than $p(1-p)$ where $p = 0.99$ were removed, eliminating baseline structural noise.
- **Random Forest Selection:** A balanced Random Forest classifier (`n_estimators=100`) was trained on the remaining features. Only features exhibiting an importance score strictly greater than the mean importance were retained for the final modeling phase.

2.3 Consensus Modeling and Hyperparameter Optimization

A weighted ensemble model was constructed utilizing Bernoulli Naïve Bayes and eXtreme Gradient Boosting (XGBoost). The XGBoost classifier was specifically regularized to prevent over-fitting on the minority class by applying a `scale_pos_weight` inversely proportional to the class imbalance. Hyperparameters for XGBoost (maximum depth, learning rate, subsample ratio, and gamma) were optimized via Randomized Search with Stratified K-Fold cross-validation, utilizing MCC as the scoring metric.

Final consensus probabilities for each compound were calculated by weighting the output probabilities of the Naïve Bayes and XGBoost models by their respective ROC-AUC scores on the test sets.

2.4 Dynamic Thresholding via MCC Optimization

To address the thresholding fallacy in imbalanced datasets, static probability cutoffs (0.50) were discarded. During training, the consensus probabilities were calculated for the training set, and the decision threshold was iteratively tested from 0.10 to 0.95 in increments of 0.05. The specific threshold that maximized the Matthews Correlation Coefficient (MCC) on the training data was dynamically selected and subsequently applied to the test data to generate final binary predictions.

3 Results

3.1 Global Pipeline Performance

The proposed methodology successfully generated customized, predictive models for all 12 Tox21 endpoints. The dual-stage feature selection effectively reduced the 1024-bit fingerprint arrays to an average of 169 to 234 highly predictive features per endpoint. The dynamic thresholding mechanism revealed that optimal classification boundaries vary drastically across biological pathways, ranging from 0.35 (NR-ER) to 0.90 (NR-AR).

The pipeline achieved an average ROC-AUC exceeding 0.80 across multiple critical targets, including NR-AhR (0.8835), SR-MMP (0.8564), and NR-Aromatase (0.8377). A selection of endpoint metrics is detailed in Table 1.

Table 1: Performance Metrics Across Selected Tox21 Endpoints

Biological Endpoint	**ROC-AUC**	**MCC**	**Opt. Threshold**	**Features Used**
NR-AR	0.7541	0.5773	0.90	195
NR-AhR	0.8835	0.4599	0.85	169
NR-ER	0.7115	0.2545	0.35	234
SR-MMP	0.8564	0.4985	0.75	189
SR-p53	0.8004	0.2671	0.80	221

3.2 Case Studies in Biological Specificity

To validate the biological accuracy of the pipeline, three compounds with well-documented toxicological profiles were evaluated across the complete 12-model suite:

- **Negative Control (Aspirin):** The model correctly predicted negative classifications across all 12 endpoints, with consensus probabilities remaining substantially below the optimized thresholds.
- **Endocrine Disruption (Bisphenol A):** The pipeline surgically identified BPA as a severe disruptor, maxing out probabilities strictly on the Estrogen Receptor pathways (NR-ER: 0.964) while correctly identifying it as safe for unrelated stress response targets.
- **Carcinogenesis (Benzo[a]pyrene):** The model successfully isolated the exact mechanistic pathways targeted by this heavy carcinogen, triggering the Aryl Hydrocarbon Receptor (NR-AhR: 0.994) and DNA-damage stress response (SR-p53: 0.971).

4 Discussion

The results demonstrate that standard machine learning approaches relying on arbitrary probability thresholds are fundamentally misaligned with the realities of high-throughput toxicological data. The variation in optimal thresholds highlights that different biological assays exhibit inherently different baseline noise levels and signal strengths.

By allowing the algorithm to define the boundary between "noise" and "activity" strictly through the mathematical lens of MCC optimization, the resulting consensus model achieves a high degree of biological selectivity. This selectivity is paramount for early-stage drug discovery, where computational models must not only flag toxic compounds but also accurately deduce their underlying biological mechanisms of action.

5 Conclusion

This study presents a highly robust, scalable computational pipeline for the prediction of in vivo toxicity from in vitro qHTS data. By integrating rigorous feature reduction, AUC-weighted consensus modeling, and dynamic MCC thresholding, the pipeline overcomes the limitations of extreme class imbalance. The resulting models demonstrate exceptional predictive capacity and mechanistic accuracy, offering a reliable, data-driven tool for predictive toxicology and computational drug design.